Computer Graphis Fluid Simulation report

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1 Clipping and Voronoï Diagrams

We call the Voronoï cell of a site P_0 the set of points that are closer to P_0 than any other site. To find this for every cell we use the Sutherland Hodgman algorithm that starts from a unit square for each cell and then clip by the bisector of $[P_0, P_i]$ for every other site P_i.

For each clip we walk over the edges of the polygon and for each vertex look if, $\langle V - M,\ P_i - P_0\rangle$, where $M = \frac{1}{2}(P_0 + P_i)$ is the bisector midpoint, is negative in which case it is kept. If an edge crosses the bisector i use this new interection point

$$t = \frac{\langle M - \text{prev},\ P_i - P_0\rangle}{\langle \text{curr} - \text{prev},\ P_i - P_0\rangle}.$$

Doing this for every P_i gives the cell of P_0, and repeating over every site gives the whole diagram. This gives an algorithm that is $O(N^2)$.

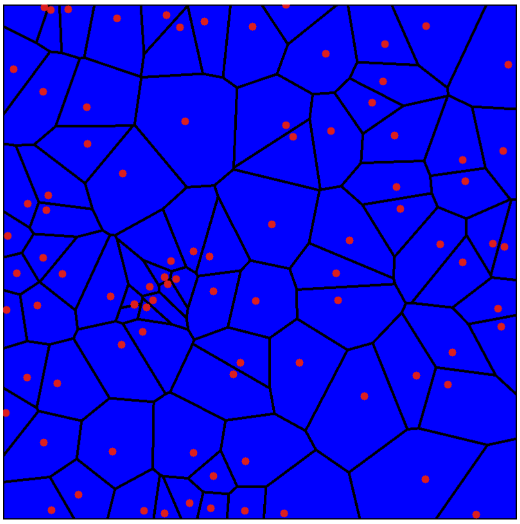


Figure 1: Voronoï diagram

2 Semi-discrete Optimal Transport (Lab 7)

Power/Laguerre cells: To control cell size we use weights w_i assigned to each site and say that a point a X belongs to the cell of P_i when

$$\|X - P_i\|^2 - w_i \leq \|X - P_j\|^2 - w_j, \qquad \forall j \neq i.$$

We can then reuuse our same vertex test and intersection formulas with our the new midpoint:

$$M' = \tfrac{1}{2}(P_0 + P_i) + \frac{w_0 - w_i}{2\,\|P_i - P_0\|^2}\,(P_i - P_0),$$

.

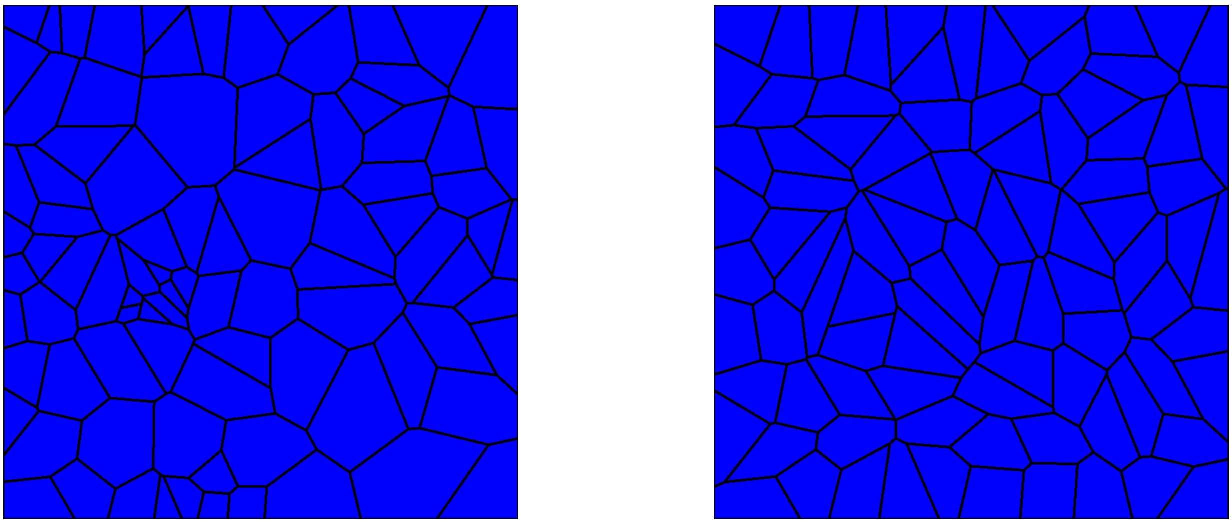
Optimising the weights with L-BFGS. We then wanted to have that each cell has a target area $\lambda = V_{\text{fluid}}/N$. This is done in our evaluate callback that rebuilds the diagram from the current weights and accumulates the objective

$$g(W) = \sum_i \left(\int_{\text{Lag}_i} \|x - P_i\|^2 \, dx \; - \; w_i \, A_i \; + \; \lambda \, w_i \right) \; + \; w_{\text{air}} \big(V_{\text{air}}^{\text{des}} - V_{\text{air}}^{\text{est}} \big),$$

where A_i is the area of cell i and $V_{\text{air}}^{\text{est}} = 1 - \sum_i A_i$. Then

$$\frac{\partial g}{\partial w_i} = \lambda - A_i, \qquad \frac{\partial g}{\partial w_{\text{air}}} = V_{\text{air}}^{\text{des}} - V_{\text{air}}^{\text{est}},$$

and since L-BFGS minimises, we pass it $-g$ and $-\nabla g$. At the optimum $\partial g / \partial w_i = 0$, so $A_i = \lambda$ for every cell.



(a) Uniform weights (b) After L-BFGS

Figure 2: 470 L-BFGS iterations

3 Computational Fluid Dynamics (Lab 8)

Disk clipping and the air variable. For the free surface we also have that each fluid cell is clipped by a circle of radius

$$r_i = \sqrt{w_i - w_{\text{air}}}$$

that is centered around P_i which I discretized as a 50 sided regular polygon and clip edge by edge as we did previously. If $w_i - w_{\text{air}} \leq 0$ the cell is empty. We optimize the air weight with the fluid weights to have that the total area $V_{\text{air}}^{\text{est}} = 1 - \sum_i A_i$ is equal to the desired air volume $1 - V_{\text{fluid}}$.

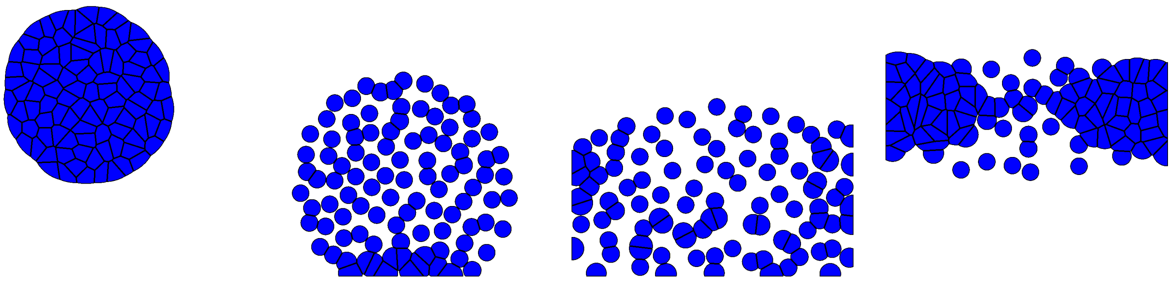
Forces and integration. Once each cell is optimised, we take the centroid C_i of each cell and apply gravity together with a spring force pulling the particle towards that centroid, so the total force is

$$F_i = m_i \, \mathbf{g} + \frac{10}{\varepsilon^2} \, (C_i - P_i).$$

This force is integrated with velocity to get position with

$$v_i \leftarrow v_i + \frac{dt}{m_i} \, F_i, \qquad P_i \leftarrow P_i + dt \, v_i,$$

using $\varepsilon = 0.004$, $dt = 0.002$ and fluid fraction $V_{\text{fluid}} = 0.20$. A particle leaving the unit square is reflected back inside. Iterating this produces the simulation frame by frame. Unfortunately my results did not produce a correct fluid simulation. As can be seen in the images above. I thought that it might be an issue with the spring force and tried to play around with spring stiffness and mass of each particle but did not get any good results. Since it seemed like thatballs bounced too high after hitting the floor I also tried to change how velocity is transferred as the particles hit the ground for example zeroing or having a dimining coefficient but these did not work either. I tried talking with the TA and my classmates but both were inconclusive. I imagine I must have made a mistake in my implementation at some point but was not able to find where. (In my main file that is on the repo I multiplied my spring force by 10 but tried with a normal spring force later on which did not solve anything.) The videos on my repo are of one with 100 balls and another of 50 balls. I used less balls as the videos were more coherent than 500 balls where the balls just bounced up and down almost not interacting with one another.



(a) $t = 0.00\,\mathrm{s}$ (b) $t = 0.16\,\mathrm{s}$ (c) $t = 0.30\,\mathrm{s}$ (d) $t = 0.46\,\mathrm{s}$

Figure 3: Four snapshort of the simulation. The full video is available in my repo.

Feedback on the course

I really enjoyed the course especially the first project. While the lecture were very dense, having the lecture notes on the side was a clear game changer. It permits you to really understand concepts you did not understand during the lecture and go into more depth on the material as well. I did feel like sometimes the course was a bit dense for a half semester course but at the same time this is a positive aspect. I really liked the first project but did think the end of the second project was a bit long especially for the last TD. This may also be due to the fact I had more difficulties during the end of the second project.